

NAME: _____

PERIOD: _____

What Did Each Simulation Estimate?

AP Statistics · Unit 2 · Topic 2.3

[STAR] THE PROBLEM

Suppose each of five students can be treated as independently selected, and each has probability 0.12 of being left-handed. Two students use 1,000 simulation trials to estimate the probability that at least 2 of the 5 students are left-handed. Their plans and results are shown.

Results: Student 1 = 87/1,000; Student 2 = 94/1,000

Plan	<i>L</i> assignment	Trial; success
1	0 of 0–9	5 digits; 2+ L
2	00–11 of 00–99	5 pairs; exactly 2 L

FIRST TAKE — one sentence before you work the parts

Which simulation result would you trust more for the question asked—Student 1, Student 2, or neither? Name one reason.

[BOOK] THREE MATCHES MAKE A VALID SIMULATION (keep on the page)

Match each outcome's chance with the fraction of equally likely labels assigned to it. One trial must reproduce the stated number of independent selections. Count the exact event named in the question. More trials reduce chance variation but do not repair a wrong assignment or event. Estimate probability with successful trials/total trials.

Work (a)–(c).

DOK 1 (a) List the left-handed counts included in at least 2 of 5. State the chance one student is left-handed.

DOK 2 (b) Find each reported estimate. For each plan, identify either a mismatch in the chance assignment or a mismatch in the event counted.

DOK 3 (c) Would you trust either plan for the question asked? Explain both mismatches and why the larger reported estimate does not validate a plan. Describe one corrected trial, including the random labels and the event counted. Write 3–4 sentences.

Frame: Plan 1 uses a chance of _____ instead of _____; Plan 2 leaves out _____.

Frame: My corrected trial uses _____ and counts a success when _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.